GEK-83858

DRIVE SYSTEM INSTRUCTIONS

KW-KVA MODULE

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC

KW - KVA MODULE

(GEK - 83858)

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FUNCTION

The KW-KVA Module is designed to prevent AC Power Plant "Blackout" by detecting excessive real (KW) and total (KVA) power loads, giving a warning and limiting the load to power actually available. It collects and distributes signals common to all the generator units and contains KW and KVA limiting and alarm circuits.

THEORY OF OPERATION

The KW-KVA circuitry performs four (4) separate functions as described below.

KW Limit

The purpose of the KW (kilowatt) limit circuit is to prevent overloading of the diesel engines. It does this by providing a signal proportional to the magnitude of the KW load. This signal is sent to all power converter regulators which then act to reduce converter output, thereby reducing engine loading.

External KW sensors (in the GEM Module) monitor the KW being supplied by each generator. The highest KW signal of all the generators on-line is selected and supplied to the input of amplifier AOAL(A). This signal is compared to a preset KW Limit bias signal from trimpot P1. If the (+) KW signal exceeds the (-) bias signal, amplifier (A) output will swing negative. This negative signal is supplied to all SCR regulators, and reduces converter output to the level required to prevent engine overload. A clamping diode, C7, prevents amplifier AOAL(A) from swinging positive. The magnitude of the KW Limit signal can be monitored at "Limit" test jack (J3). A typical value is -4.8 volts for 650KW Limit. The KW Limit signal is adjustable via trim pot P1 to accommodate various diesel engine horsepower (KW) ratings.

KVA Limit

The purpose of the KVA limit circuit is to protect the AC generator from overload. It does this in a manner similar to that for KW limit described above. Amplifier (D) accepts the highest KVA input via inverting amplifier AOAL(C) and trimpot P3 sets the desired KVA bias limit. The KW limit and the KVA limit signals are diode OR'ed (diodes CR1 and CR2). This ensures that either an excessive KVA load or an excessive KW load will reduce the power converter output.

THEORY OF OPERATION (Continued)

High Load Indication (Provides Relay Interlock Output)

The purpose of this circuit is to indicate, by means of a relay interlock, whether the system is approaching or is in KW or KVA limit. The negative KW-KVA limit signal is applied to a comparator gate unit AOAL(G). When this input becomes zero (IN 0V) the output will go to +5 volts. This signal is applied to amplifier AOAL(F) - a high gain power amplifier. Its output will swing negative to energize reed relay HLR whose interlock is used to indicate HIGH-LOAD on the operator's console. This interlock is generally set to close five percent (5%) lower than the actual limit value so as to provide the operator an "early warning" prior to actually limiting converter output.

KW Totalizing (If used with operator console mounted meter)

Individual signals proportional to each generator's KW load are summed in amplifier AOAL(B). Its output drives the "%KW" meter on the operator's console (if provided). The meter is divided into segments, one for each of the available engines, and is calibrated (trimpot P2) so that each segment represents 100% KW load per engine. This allows the operator to judge the degree of loading versus the number of engines on line. For example with four engines, each input to AOAL(B) would be equal to 2.5V at full output with each 2.5 volts representing a segment of the meter "pie", which has four (4) segments. Meter movement is 0 to 10VDC, such that with four fully loaded engines, we have 4 X 2.5 = 10VDC and the meter reading at the end of the fourth segment.

With one (1) engine generator on line and the meter reading near the end of the first segment, another engine generator should be readied for placing on the line.

ELEMENTARY MNEMONICS

The following mnemonics or abbreviations are used in the elementary shipped with your equipment. Their meanings are explained below.

Device Designations

AOAL - Analog Operational Amplifier; there are five separate operational amplifiers, "A" through "F", plus comparator circuits.

C__ - Identifies a Capacitor

CR__ - Identifies a Diode

GEN__ - Generator Input to KW-KVA Module

HLR__ - Identifies a High Load Relay

J1 - J7 - Test Jacks for measuring various signal levels

P__ - Identifies a Potentiometer

R__ - Identifies a Resistor

TBOA - 12-Point Terminal Board for KW and KVA Signal Input

TBOI - 12-Point Terminal Board for KW Signal Output

Signal Designations

ACOM - Analog Common. Common connection for signals, (0 ± 15V) input to the module (except for high level signals) (TBOA pin #11 and TBOI pin #8)

ACOM-GC - Analog Common - Generator Control. Common connection for signals input to module from the generator-engine module (TBOA pin #2)

DKVA - Diode KVA signal indicating highest value of generator output. (Also see KVA signal designation.) Negative Input signal to module. (TBOA Pin #3)

DKW - Diode KW signal from generators which represents the largest output being generated. (Also see KW + signal designation.) Positive Input signal to module. (TBOA Pin #1)

ELEMENTARY MNEMONICS (Continued)

GEN # KW - Positive KW Limit signal for Generator number # KW. (Also see KW - # signal designation.) Input to KW Totalizer Operational Amplifier Summer. (TBOA pins 4, 5, 6, and 7)

HLR - High Load signal to indicating light on the operator's console (TBOA pins 8 and 9 - if used)

KVA+ - Positive KVA signal indicating highest value of generator output. (Also see DKVA signal designation.) Output from module. (TBOI pin #4)

KW/KVA - Kilowatt/Kilovolt-Amperes limit signal to PCP interface module. (TBOI Pin #2)

KW- - Negative Generator # KW limit signal. (Also see GEN # KW signal designation.) Output from module. (TBOI Pin #1)

KW+ - Positive KW signal from generators which represents the largest output being generated. (Also see DKW signal designation.) Output from module. (TBOI Pin #3)

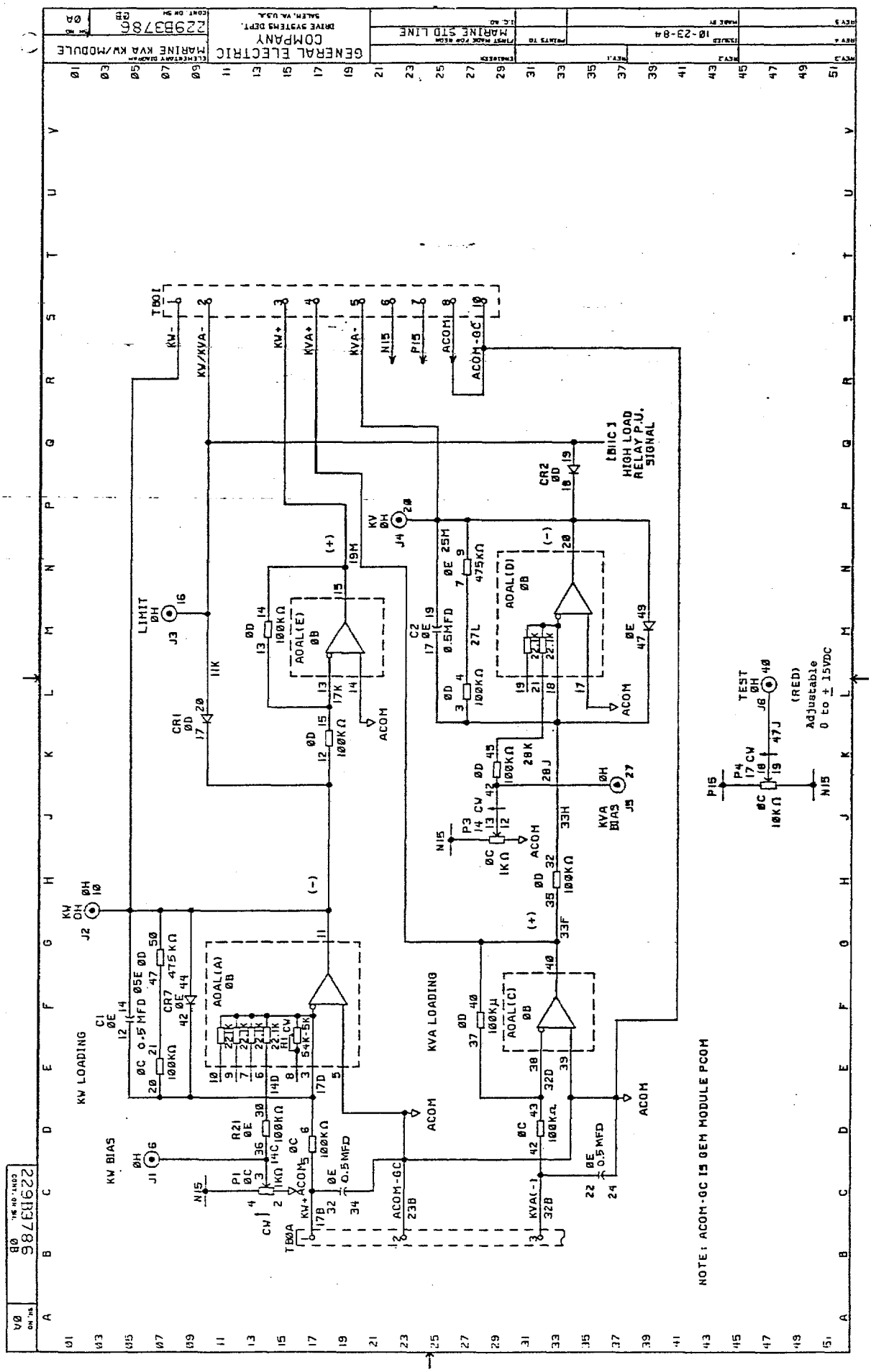
KVA- - Negative KVA signal indicating highest value of generator output. (Also see DKVA signal designation.) Output from module. (TBOI Pin #5)

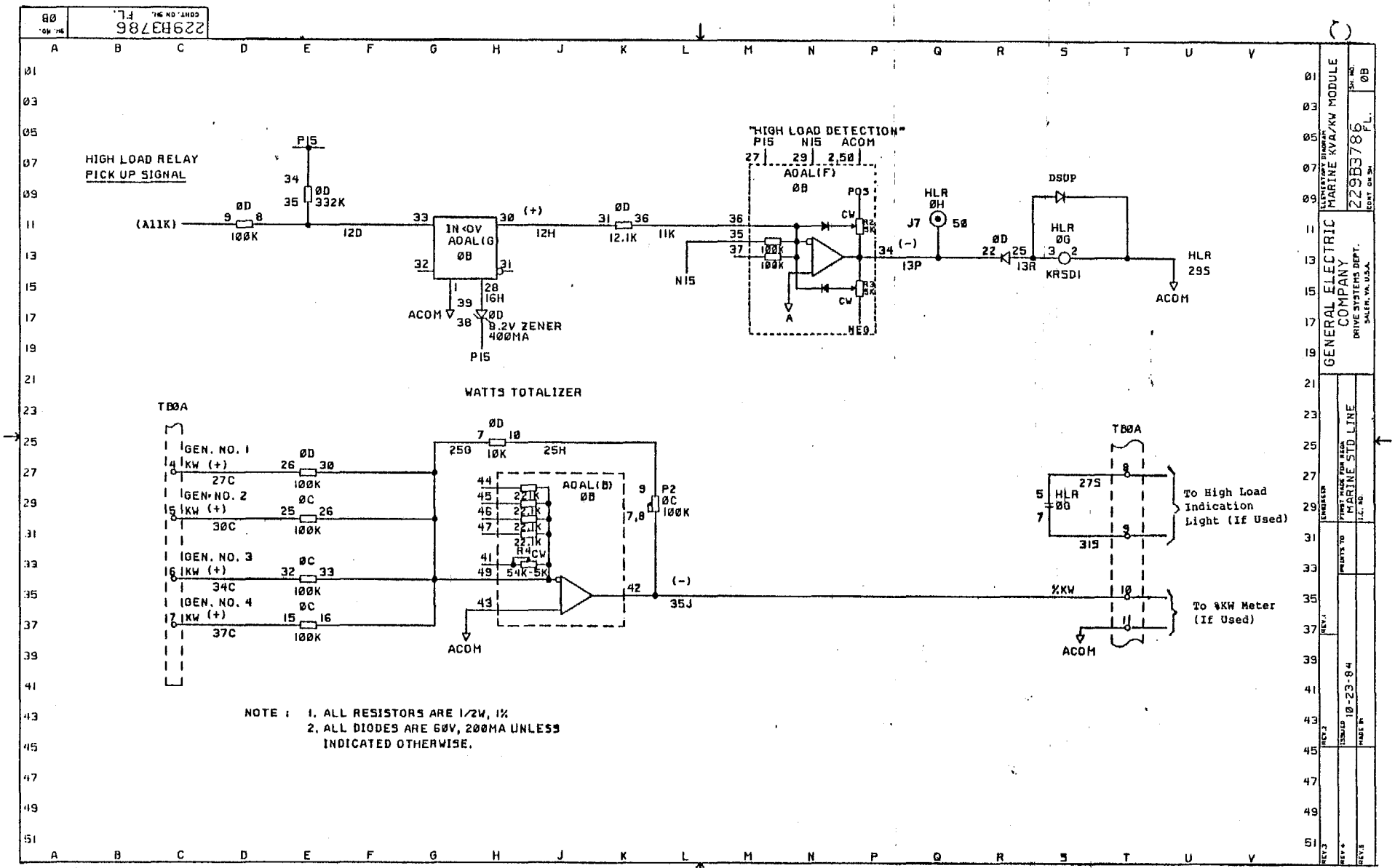
N15 - Negative 15 volt DC bus (TBOI Pin #6)

PCOM - Peripheral common (TBOA Pin #2 connects to module ACOM)

P15 - Positive 15 volt DC bus (TBOI Pin #7)

%KW - Negative output percent KW signal to meter on operator's console. (TBOA Pin #10 - if used)


GENERAL ELECTRIC COMPANY
MARINE KVA KW/MODULE
MARINE STD LINE
10-23-84
NOTE: ACOM-GC IS GEM MODULE PCOM
KW LOADING
KVA LOADING
KW BIAS
KVA BIAS
LIMIT
TEST
HIGH LOAD RELAY P.U. SIGNAL
(RED) Adjustable 0 to ± 15VDC
ACOM
ACOM-GC


HIGH LOAD RELAY
PICK UP SIGNAL
(A11K)
100K
12D
IN<0V
AOAL(G)
0B
12H
12.1K
11K
"HIGH LOAD DETECTION"
P15 N15 ACOM
AOAL(F)
0B
N15
13P
HLR
0H
J7
50
DSDP
HLR
0G
KRSDI
13R
HLR
29S
ACOM
8.2V ZENER
400MA
16H
WATTS TOTALIZER
TB0A
GEN. NO. 1
KW (+)
27C
GEN. NO. 2
KW (+)
30C
GEN. NO. 3
KW (+)
34C
GEN. NO. 4
KW (+)
37C
100K
25D
25H
AOAL(B)
0B
P2
0C
35J
(-)
ACOM
27S
HLR
0G
31S
To High Load
Indication
Light (If Used)
%KW
To %KW Meter
(If Used)
NOTE : 1. ALL RESISTORS ARE 1/2W, 1%
2. ALL DIODES ARE 60V, 200MA UNLESS
INDICATED OTHERWISE.
GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
MARINE KVA/KW MODULE
229B3786 FL.
0B
10-23-84
MARINE STD LINE

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SALEM, VIRGINIA 24153

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